Predictive Analytics :Understanding Customer Behavior Through Machine Learning (Unsupervised Learning Algorithms)

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Introduction

A look into unsupervised machine learning algorithms that are beneficial for business establishments. Find out how companies harness the power of predictive analytics to maximize their profits and decrease loss. We discuss a wide range of unsupervised machine learning algorithms that cover customer segmentation, dimension reduction, and anomaly detection.

K-Means Clustering

Predictive analytics uses data, statistical algorithms, and machine-learning techniques to predict trends, improve performance, encourage decision making, and predict behavior based on historical data.
When predicting customer behavior, mapping historical data into behavioral data is the main objective at the preprocessing stage, and by using predictive modeling algorithms, behavior pattern analysis will identify patterns of customer behavior and then build predictive models, which will eventually be used to predict results from new data.

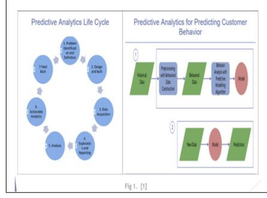


Fig 1. [1]

Customer segmentation is the process of dividing the customers into groups which shares similar characteristics, and its purpose is to determine how to deal with customers in each category.

Fig 2. Applying K-Means algorithm to Customer Segmentation [2]

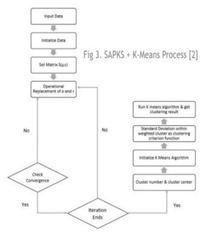


Fig 3. SAPKS + K-Means Process [2]

In K-Means clustering, all variables should be normally distributed and have the same variance. In addition, all clusters have the same number of observations, and that there's little to no correlation in the relationship between variables.
To demonstrate how K-Means algorithm works, the data set shown in figure #6 will be used. Let's say that there are some new products on the market that you are interested in selling and you want to target a specific type of clients for each one of the products.

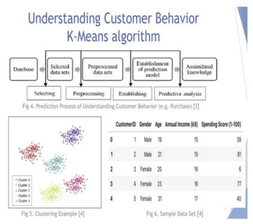


Fig 4. Prediction Process of Understanding Customer Behavior (e.g. Purchase) [3]

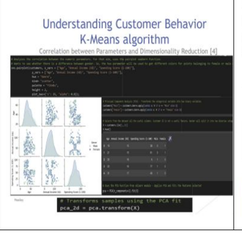
Fig 5. Clustering Example [4]

CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	Male	19	15	39
1	Male	21	15	81
2	Female	20	16	6
3	Female	23	16	77
4	Female	31	17	40

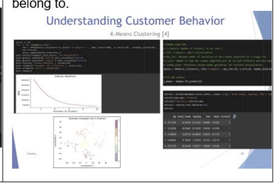
Fig 6. Sample Data Set [4]

After reading the data into a DataFrame, we start exploring this data and checking its quality by making sure there's no duplicates or missing values.
The next step is to observe the distribution of the variables. Two functions will be used for this purpose. The first one will retrieve the descriptive statistics of the variables, and the second one will help graph the variables distribution.

The correlation between the parameters is analyzed to help visualize a summary of the data.
After checking that K-Means can be applied, Principal Component Analysis (PCA) can be used to discover which dimensions best maximize the variance of features involved. During the fitting process the model learns the "components" and the "explained variance" and the samples are transformed using the PCA fit.



K-Means Clustering typically uses Euclidean distance to determine how similar 2 points are but, first it is required to fix the number of clusters to use which can be done by employing the Elbow method.
The main idea behind K-Means is to select k centers, one for each cluster. Then, it calculates the Euclidean distance between each point and the cluster centers. It assigns the points to the cluster center where the distance is minimum and repeats the process until the centers do not move anymore.
As you can see, the data has been clustered into 5 segments. This model can be used to predict which segments new customers will belong to.



Density Based Clustering

The DBSCAN algorithm works by :
- First, setting two parameters: Epsilon and Min_Samples
  - Epsilon: sets the radius from the central point to find the nearest neighbor
  - Min_Samples: sets the amount of data points needed within the "neighborhood" to be considered part of the cluster (also called the density threshold)
- Then, pick a random data point - called the central point
- From the central point, consider
  - Are there Min_Samples amount of data points within the radius?
    - If not
      - That central point is not part of the cluster
    - Else
      - The central point is part of that cluster
- Once all neighboring points are checked
  - Move on to another random point
- Repeat above steps

Why consider DBSCAN for a viable clustering algorithm?
- The algorithm clusters "noise" – which are outlier data points
- DBSCAN finds associations that would be hard to find manually

The process of exploring DBSCAN:
- Implement the DBSCAN algorithm
- Test different amounts of Min_Samples: 4,6,8
- Question: How does one determine the best amount of Min_Samples for clustering?
  - Discovered that making a Nearest-Neighbor graph helps you find the best amount for Min_Samples
  - However, when the density threshold is X, there are clusters that overlap, which shouldn't happen
- Explore density thresholds
  - Take a deep dive into the paper "An Improved DBSCAN Algorithm Based on the Neighbor Similarity and Fast Nearest Neighbor Query"
    - Start the implementation of this method to:
      - Understand how to find a meaningful density threshold
      - Attempt to improve time efficiency

Whats next?
- To determine if improved DBSCAN is a reliable option, it should have a better time efficiency and be able to handle large data sets

Graph of Mall_Customers Dataset using DBSCAN

Locally Linear Embedding

sLocally Linear Embedd - LLE
Locall Linear embedding is a method of dimensionality reduction it makes the non liner graph to be linear graph.
1) it finds neightvbors of x point and calculate the weights with number of neighbors with the weights it starts to calculates the point of y.

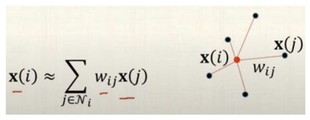


figure1 - Formula for getting weights

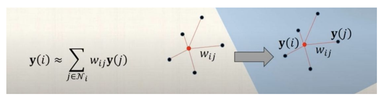


Figure2 - How to get y out of weights

```
params = {
    "n_neighbors": n_neighbors,
    "n_components": n_components,
    "eigen_solver": 'auto',
    "random_state": rng,
}
lle_standard = manifold.LocallyLinearEmbedding(method='standard', **params)
S_standard = lle_standard.fit_transform(S_points)

lle_ltsa = manifold.LocallyLinearEmbedding(method='ltsa', **params)
S_ltsa = lle_ltsa.fit_transform(S_points)

lle_hessian = manifold.LocallyLinearEmbedding(method='hessian', **params)
S_hessian = lle_hessian.fit_transform(S_points)

lle_mod = manifold.LocallyLinearEmbedding(method='modified', modified_tol=0.9, **params)
S_mod = lle_mod.fit_transform(S_points)

fig, axs = plt.subplots(
    nrows=2, ncols=2, figsize=(7, 7), facecolor='white', constrained_layout=True
)
fig.suptitle('Locally Linear Embeddings', size=16)

lle_methods = [
    ('Standard locally linear embedding', S_standard),
    ('Local tangent space alignment', S_ltsa),
    ('Hessian LLE', S_hessian),
    ('Modified locally linear embedding', S_mod),
]
for ax, method in zip(axs.flat, lle_methods):
    name, points = method
    add_2d_scatter(ax, points, S_color, name)

plt.show()
```
Figure 3 - Code for LLE variants

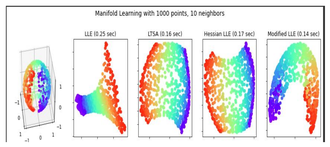
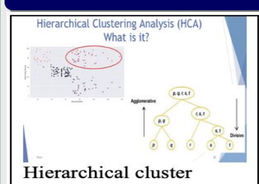


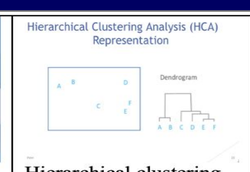
Figure 4- Result of LLE variants

cFigure 3 is showing how the Local Linear Embedding is being processed. first it needs number of neighbors that will be used to solve for weights and then it takes in the number of components that will be resulted out as y. in this code 1 standard LLE and 3 LLE variantants and the figure 4 is showing the result of code. Left most one in the figure 4 is original graph that is sphere and each linear graph ahs name tag of which LLE is being used. all of them might have different linear graph but it still has the characteristics from original sphere graph and successfully made non-linear graph to linear graph.

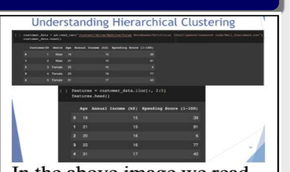
Hierarchical Clustering Analysis



Hierarchical cluster analysis is an unsupervised clustering algorithm which involves creating clusters that have predominant ordering from top to bottom. The most common type of hierarchical clustering is agglomerative hierarchical clustering which is a bottom-up approach. The divisive hierarchical clustering is a top-down clustering method.

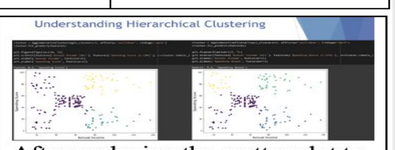


Hierarchical clustering is represented using Dendrograms They represent the hierarchical relationships between objects. Y-axis is the distance between clusters and X-axis is the order of clustered groups.



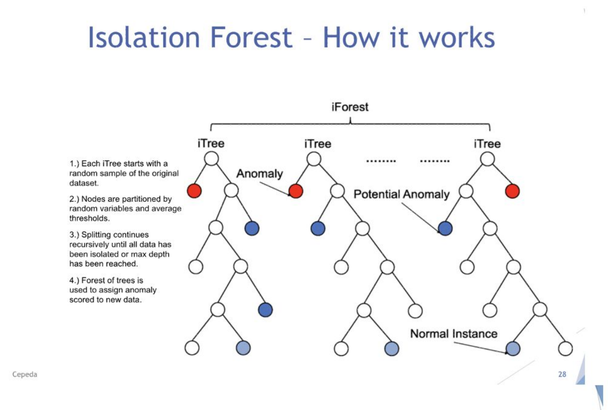
In the above image we read the dataset into a data frame and output the first 5 lines of code. Before we begin to apply hierarchical clustering to the data set, we must change the data frame to only include the relevant columns which are age, annual income, and spending score as shown in the bottom image.

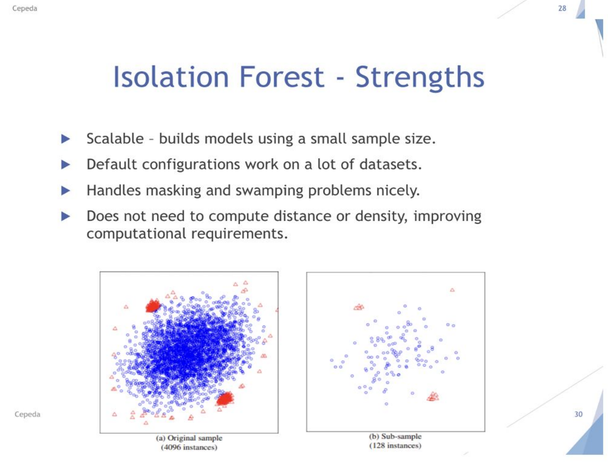
To make a dendrogram we must import the scipy.cluster.hierarchy as shc and link the data frame with the relevant columns. Three seems a suitable threshold as per the rule for cutting for finalizing the optimal number of clusters. In the image to the right, we do the cut when y = 300 and that makes it to where there are supposed to be 3 clusters in the scatter plot.



After analyzing the scatter plot to the left we can see that there is a need for a 4th cluster to separate a low and high spending score for people with low incomes. After analysis of the plot to the right with 4 clusters we can say that we are interested in the green-colored cluster because these customers have a high annual income along with a high spending score and that is what is being looked for in this customer segmentation.

Anomaly Detection Isolation Forest





(a) Original sample (4096 instances)
(b) Sub-sample (128 instances)

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